UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Chapter 1- Tutorial

SET THEORY

[Part 1: Set & Subset]

Premier Digital Tech University™

Exercise

Answer true or false

a) $\{x\} \subseteq \{x\}$

b) $\{x\} \in \{x\}$

c) $\{x\} \in \{x,\{x\}\}$

d) $\{x\} \subseteq \{x,\{x\}\}$

e) $\{\{x\}\} \subseteq \{x,\{x\}\}$

f) $x \in \{x,\{x\}\}$

Exercise

Determine whether each pair of sets is equal

a) {1, 2, 2, 3}, {1, 3, 2}

b) $\{ x \mid x^2 + x = 2\}$, {1, -2}

c) $\{x \mid x$ is a real number and $0 < x \leq 2\}$, {1,2}

Exercise

If M is finite, determine the $|M|$

a) $M = \{1, 2, 3, 4\}$
b) $M = \{4, 4, 4\}$
c) $M = \{\}$
d) $M = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$

Exercise

List the member of $P(\{a, b, c, d\})$.
Which are proper subset of {a, b, c, d}?

Exercise

a) Let $A = \{a, \{a\}, b, \{a, b\}\}$. State whether the following statement is TRUE or FALSE.

i. $\{a\} \subseteq A$ (1 mark)

ii. $\{a, b\} \in A$ (1 mark)

b) Given set $B = \{x | x \in Z, x^2 + 6 \leq 10\}$, where Z is the set of integers. List the elements of set B. (1 mark)

Exercise

a) Calculate the size (cardinality) of these sets, given $A=\{1, 3, 5\}$ and $B=\{2, 4, 6, 8\}$. This can be done using only the facts that $|A| =3$ and $|B| =4$. You do not need to list all elements of the new sets.

i. $|A \times B|$

ii. $|P(A) \times P(B)|$

(2 marks)